

sloopDX Cheat Sheet

DX7 FM synth + FM drums for the M-VAVE FM-1
Version 2.1 · dx7.designburgapps.com · github.com/zvenson/dxsloop

THE FOUR TRACKS

1 2 3 = DX7 synths (8 voices shared) · 4 = drums (6 voices of their own). Every track has its own pattern.

BASICS

ALGORITHM	select the track (on every page)
PRESETS	sound: 01–20 factory, 21–52 your bank, then user presets · drums: the kit
SELECT	tempo
KNOB 1–4	HOME: BRITE · ATK · REL · FDBK of the voice (act at once)
PLAY	start / stop all tracks
REC	playing: record now · stopped: armed, the first note starts · empty project: play freely, REC on the "1" closes the loop
hold REC	clear the track (a ring fills, ~2 s)
OCT– / OCT+	octave (both = 0) · drums, held: ghost / hard hits
HOME	TRACKS · hold: menu (colour, low cut, lights, USB audio, about)
EDIT + OCT–/+	undo / redo

LAYERS: HOLD A BUTTON

Hold	White keys	KNOB 1 · 2 · 3 · 4
FX	punch-in effect	filter · dust · duck
EDIT	erase that sound / note	shift · length ×2½ · transp.
ARP	note repeat (roll)	rate
SEQ	steps 1–16 · drums, step held: KNOB 4 tune lock, PRESETS decay lock (★)	sound/note · div · swing · length
SCL	the key of the song	chord · scale · keys · transp.
GLO	1–4 mute · 5–8 solo · 16 tap	level of tracks 1–4
SAVE	1–4 play section A–D · 5–8 store · 13 loop/song · 14 song rec	

Lock a layer: hold its button + tap **HOME**. Tap instead of hold: its pages.

EDIT A DX7 VOICE: TAP EDIT

SELECT	choose the row
ALGORITHM	change the value (the first turn on "Algorithm" only draws it)
EDIT	open a group / run an action (tap twice)
HOME	one level back
PRESETS	straight to the next / previous operator
KNOB 1 · 2	CUTOFF · RESONANCE of the low-pass (the value shows at the bottom) · KNOB 3 · 4 as on HOME
SAVE	store the bank (a dot at the bottom = not stored)

List: Voice · Bank 1–8 · Algorithm 1–32 · Feedback · Osc Sync · Transpose · **OP1–OP6** > · Pitch Env > · LFO > · Name > · Copy To · Init Voice · Store · More Pages >

Operator: On · Output Level · Coarse (ratio) · Fine · Detune · Osc Mode · Rate 1–4 · Level 1–4 · Key Velocity · Amp Mod Sens · Break Point · Left/Right Depth · Left/Right Curve · Rate Scaling

Changing a factory voice copies it into the first free user slot first ("COPIED TO U01"). In the algorithm drawing: carriers cyan, modulators light blue.

SHAPE THE SOUND

BRITE	the modulators' level: dull ↔ bright
ATK · DEC · REL	slower / longer (0 = as programmed)
FDBK	feedback: sine → saw
CUT · RESO	a low-pass behind the voice (CUT 127 = open, out of the way)
ENV page	ATK · DEC · REL of the carriers, SUS lowers the sustain (127 = as programmed)
ENV/LFO DEST	FLT = the cutoff, SHP = brightness, PIT = pitch, AMP = volume

DRUMS: TRACK 4, WHITE KEYS

F3	G3	A3	B3	C4	D4	E4	F4
Kick	Kick 2	Snare	Clap	Hat	Open	Pedal	Rim
G4	A4	B4	C5	D5	E5	F5	G5
Snare 2	Tom L	Tom H	Crash	Ride	Shaker	Conga	Cowbell

Kits (PRESETS): 1 DX KIT · 2 808 FM · 3 ELECTRO · 4 METAL · 5 **MY KIT**. A black key plays the sound left of it. **SEQ**: grid ↔ kit. **Kit page** KNOB 1–4: kit · level · **drive** · **comp**. **EDIT**: the **lane** (the sound last played): TUNE · DECAY · SWEEP · BRIGHT, **PRESETS**: NOISE · LEVEL · PAN · CHOKE, then REV. **EDIT** 2×: the **dice** (seed top right). **SAVE** 2×: store as MY KIT.

YOUR OWN DX7 BANKS (.SYX)

1. Chrome / Edge: **dx7.designburgapps.com** → Editor → **Connect**.
2. Tab **Library**: pick "Bank in use" 1–8, drop the .syx (32 voices, 4104 bytes).
3. On the FM-1: PRESETS 21–52, or EDIT → Voice. Switch banks: EDIT → Bank.

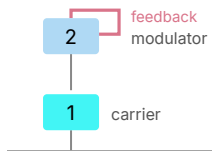
8 banks × 32 = 256 voices in memory. Original DX7 ROMs: yamahablackboxes.com · collections: bobbyblues.recup.ch

SAVE & RESCUE

Project	tap SAVE → PROJECT: slot, turn twice = save / load (4 slots, autosave)
Sound	SAVE → USER: 32 user presets
Backup	editor → Projects → Backup (projects, presets, DX7 bank)
Update	dx7.designburgapps.com → Install (USB straight in, no hub)
Go back	M-UPGRADE + the original firmware · emergency: FM-1-transporter

OPERATORS

A DX7 voice is made of **six operators**. Each is a sine oscillator with its own **envelope** and **level**. An operator is either a **carrier** (you hear it) or a **modulator** (it bends the wave of the operator below it: the more level, the more harmonics). That is FM: the brightness comes from the modulators. sloopDX adds a low-pass behind each voice (CUT, RESO) for the classic synth sweep.



Operator 2 modulates operator 1. Light blue = modulator, cyan = carrier, the pink loop = **feedback** (an operator modulating itself: from sine to saw to noise).

ALGORITHMS 1-32

The **algorithm** says how the six are wired: stacks (2 → 1, deep and bright), pairs (5: three electric-piano-like pairs), one modulator on several carriers (22), or all six as carriers (32: an organ of sines). On the FM-1: EDIT → *Algorithm*, the first turn draws it, the next ones change it.

REACHING THE OPERATORS

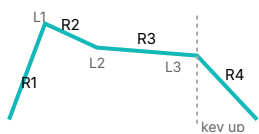
1. Synth track (1-3): tap **EDIT**. The voice list opens.
2. **SELECT** to **OP1 ... OP6**, tap **EDIT**: that operator's rows.
3. **PRESETS**: the next / previous operator, on the same row (compare Rate 1 of all six quickly).
4. **ALGORITHM** changes the value, **HOME** goes back, **SAVE** stores.
5. Row **On**: switch the operator off to hear what it does (not stored).

In the web editor (tab **Voice**) all six are on one page, with their envelopes drawn.

FREQUENCY: COARSE, FINE, DETUNE

Coarse	the ratio to the note: 0.50, 1, 2, 3 ... 31. Whole ratios on a modulator = harmonic (organ, brass, bass); 1.41, 3.5 ... = inharmonic (bells, metal)
Fine	between the ratios (1.00 → 1.99)
Detune	-7...+7: two carriers slightly apart beat (chorus, e-piano shimmer)
Osc Mode	FIXED: a frequency regardless of the key (1, 10, 100, 1000 Hz ranges): noises, drums, clicks

ENVELOPES: 4 RATES, 4 LEVELS



Rates are **speeds** (99 = instant, low = slow), levels are **targets**. R1→L1 attack, R2→L2 and R3→L3 decay to the sustain L3 (held while the key is down), R4→L4 release. A carrier's envelope = volume, a modulator's = brightness over time.

TOUCH AND THE KEYBOARD

Key Velocity	0-7: harder = louder (carrier) or brighter (modulator)
Rate Scaling	0-7: high notes decay faster, as a piano
Break Point, Depth, Curve	level up or down the keyboard from a key: thinner bass, softer treble
Amp Mod Sens	how much the LFO's AMD reaches this operator (tremolo)

PITCH ENVELOPE AND LFO

Pitch Env: 4 rates, 4 levels for the pitch of all operators (level 50 = no shift): a blip on the attack, a slow drop. **LFO**: wave (triangle, saw, square, sine, S&H), speed, delay, **PMD** (vibrato) with Pitch Mod Sens, **AMD** (tremolo) with Amp Mod Sens, key sync.

FM DRUMS: THE LANE PAGE

Each drum is a DX7 voice too: a **body** (a carrier, swept down at the hit), modulators for **tone**, and **noise operators** (drums only: noise instead of a sine, its frequency the colour). On the kit page tap **EDIT** for the sound you last played: **TUNE** the note, **DECAY** every falling rate, **SWEEP** the pitch drop, **BRIGHT** the modulators, **NOISE** the noise, then **LEVEL**, **PAN**, **CHOKE** and the **REV** send. Locks put a TUNE / DECAY on one hit.

RECIPES: START FROM 20 INIT VOICE

Bass	Alg 1 or 5 · OP2 Coarse 1, Level 70-85, fast decay (R2 ~70, L2 0) · OP1 carrier · Feedback 4-6 for growl
Sub / 808	OP1 alone at Coarse 0.50: a pure sine an octave down · 808: Pitch Env L4 66, R1 72 (it drops into the note), L3 0 (it dies away)
E-piano	Alg 5 · carriers Coarse 1 with Detune ±2 · one modulator Coarse 14 with a short decay (the tine), Key Velocity 5-7
Bell	Alg 5 · modulator Coarse 3.5 (Fine) Level 70 · carrier long R3/R4 · L3 0 (it rings out)
Brass	Alg 22 · modulator envelope with a slower R1 than the carriers (the tone opens), Feedback 7, a little Pitch Env attack
Organ	Alg 32 · six carriers at Coarse 0.5, 1, 2, 3, 4, 6 (the drawbars), Levels set the mix
Pad	slow R1 (30-45) on the carriers, long R4, two carriers detuned ±3, LFO PMD a little

Then: **Copy To** a slot, **Name** it, **SAVE**. Factory voices are copied into a user slot automatically on the first change.